

Data Management Plan

Large Medical Group Data Project

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Module 10 Summative Assessment - Data Ethics

The data project is managing health care information which includes sensitive data such as Personally Identifiable Information (PII) and Protected Health Information (PHI), and must comply with HIPAA regulations and internal policy. For a successful project result, this plan includes cleaning and identifying all sensitive data sets, including identity, access, users, and roles, as set forth below.

DATA TYPES COLLECTED

The types of data are divided into three categories: PII, PHI, and Operational data.

Personally Identifiable Information (PII)

1. Social Security Numbers (SSNs)
2. Full names
3. Dates of birth
4. Addresses

Protected Health Information (PHI)

1. Diagnoses, treatment history, and procedures
2. Medical record numbers
3. Insurance details
4. Appointment and encounter history

Operational Data

1. Billing codes
2. Provider assignments
3. Facility and scheduling data

ACCESS TO DATA SETS

Access to the data sets are divided into three functional groups: Data Analysts; Supervisors/Project Leads; and IT and Security Teams.

1. **Data Analysts:** Clean, process, and analyze sensitive data.
2. **Supervisors/Project Leads:** Review data processes and ensure compliance.
3. **IT and Security Teams:** Manage infrastructure, access control, encryption, and backups.

USERS OF THE DATA

Users are put into the following four data utilization categories: Clinical Staff; Administrative Staff; Compliance Officers; Executives and Stakeholders.

1. **Clinical Staff** (Doctors, Nurses, Technicians): Access patient health records for treatment.

2. **Administrative Staff** (Schedulers, HR, Billing): Use data for appointments, payroll, and insurance claims.
3. **Compliance Officers**: Audit data usage and privacy compliance.
4. **Executives and Stakeholders**: Use aggregated data for strategy and reporting.

USER ROLES

Six user roles are specified below in Table 1: User Role & Purpose of Data Use

Table 1: User Role and Purpose of Data Use

User Role	Purpose of Data Use
Doctor/Nurse	Review patient history for diagnosis/treatment
Scheduler	Coordinate appointments with appropriate specialists
Accounts Payable	Process billing, insurance claims
HR	Verify employee eligibility or compliance
IT Security	Manage secure access and data storage
Data Analyst	Clean, merge, and analyze datasets

DATA ROLES

Data roles are separated into four functional groups: Data Engineer; Data Analyst; Compliance Officer; Medical Admins/Staff. See Table 2: Data Roles

Table 2: Data Roles

Role	Function
Data Engineer	Handles ingestion and secure storage of sensitive data
Data Analyst	Performs data cleaning and quality checks before analysis
Compliance Officer	Ensures data use aligns with HIPAA and internal policy
Medical Admins/Staff	Access individual records for scheduling and clinical support

POSSIBLE ROLE CHANGES OVER TIME

There are four possible factors that would require role changes: Regulatory or Policy Changes; Scanning Tools; Audits; and De-identification.

Regulatory or Policy Changes

As regulatory requirements (i.e. HIPAA updates) or internal policies evolve, new compliance roles may emerge.

Scanning Tools

Machine learning models or decision support tools may require access to anonymized aggregated data.

Audits

External auditors or insurers may be granted temporary access under secure protocols.

De-identification for research

De-identification for research or long term studies may be needed in the future.

DATA DOCUMENTATION

Metadata Standards

FHIR - for healthcare-specific data exchange and interoperability, Fast Healthcare Interoperability Resources (FHIR) is the preferred standard and will be used.

Custom Data Dictionaries

Custom data dictionaries will be required to describe local data models.

1. Field names and descriptions
2. Data types and formats (e.g., SSN as string, DOB as YYYY-MM-DD)
3. Valid ranges or categorical values
4. Data provenance (source, collection date)

Additional Documentation

Other operating documents which are needed: Data access logs; ETL process documentation; Compliance documentation; and Standard Operating Procedures (SOPs).

1. **Data access logs** (for audits; who accessed what and when)
2. **ETL process documentation** (data flow, transformation steps)
3. **Compliance documentation** (e.g., HIPAA risk assessments)
4. **Standard Operating Procedures (SOPs)** for data handling and redaction

DATA QUALITY

Cleaning and Validation

Before storage, Data should be cleaned and quality-checked before long-term storage or analysis.

Important Steps:

1. Format validation (e.g., 9-digit SSNs, valid DOB ranges)
2. Duplicate detection (e.g., same patient entered multiple times)
3. Missing data checks (e.g., mandatory fields like SSN or diagnosis codes)
4. Consistency checks (e.g., matching insurance provider names and codes)
5. Insure audit trail logging for all changes

Tools & Practices:

1. Use automated validation scripts (I.e. Python with SQL)
2. Apply de-identification or tokenization where appropriate
3. Implement version control for datasets and cleaning scripts

SUMMARY

Careful adherence to data management plan for Large Medical Group will ensure data is properly managed, cleaned, and documented while maintaining compliance, security and ease-of-use for the organization and all it's users and roles. It will guide to a successful project result. If there are any suggestions for improvement to the plan or advice you have, please schedule time or notify me in Slack or by email.